

Applications of Queues

Q. Discuss the significance of Queues in computer science. Elaborate on various real-world and system-level applications where the Queue data structure is essential.

[DEF] Definition to Remember

A Queue (First In, First Out) is fundamentally used as a buffer in scenarios where a resource is shared among multiple consumers, or when data is transferred asynchronously between components operating at different speeds. It ensures fairness and order of arrival.

1. System-Level Applications (Operating Systems & Networks)

Application	Description
CPU Scheduling	OS manages processes waiting for CPU t
Disk Scheduling	I/O requests to the hard disk are queue
Spooling (Print Buffer)	Multiple documents sent to a printer a
Network Routers	Data packets arriving at a router are

2. Real-World & Algorithmic Applications

Application	Description
Call Center Routing	Customer calls are queued and routed t
Breadth-First Search (BFS)	Graph traversal algorithm uses a queue
Messaging Systems	Asynchronous systems (like Kafka, Rabb

3. Print Spooler Example Diagram

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[User 1: Doc A]
[User 2: Doc B]    [ A | B | C ]    [ Printer ]
[User 3: Doc C]    Print Queue
                    (A prints first)
  
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Because components run at different speeds (fast CPU vs slow printer), the queue temporarily holds data until the slower component is ready.

[MUST-WRITE] Must-Write Points (for 10 marks)

- Queues act as buffers to synchronize components operating at different speeds.
- They ensure fairness by preserving the order of arrival (FIFO).
- CPU Scheduling: OS uses a Ready Queue (e.g., Round Robin) to assign CPU time.
- Spooling: Printers use a print queue to handle multiple document requests in order.
- Network Routers: Store incoming packets in a queue to prevent data loss.
- BFS (Breadth-First Search): Graph algorithm uses queues to track nodes to visit.
- Used in real-world scenarios like Call Centers and Message Broker systems (Kafka).

[RECALL] Quick Recall

Queue Applications -> Act as Buffers -> OS (CPU Scheduling, Disk I/O) -> Hardware (Print Spooling, Network Routers) -> Algorithms (BFS in Graphs) -> Real-world (Call centers, Message Queues)